

# Emotion-Preserving Music Recommendation from Autobiographical Narratives for Older Adults

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**Abstract**—Social isolation among older adults is associated with cognitive decline and reduced quality of life. Music strongly evokes autobiographical memories and emotional responses, yet existing recommendation systems do not account for the emotional content of personal narratives. This study presents MelodyMemo, a computational framework that extracts emotions from autobiographical stories using natural language processing and recommends emotionally aligned music for older adults. The system consists of two modules. **R1 (Emotion Extraction)** applies a pretrained DistilRoBERTa classifier to analyze autobiographical stories, infer emotional categories, and map them to musical attributes (tempo, valence, energy). **R2 (Music Recommendation)** takes the derived attributes and generates song recommendations using collaborative filtering, content-based filtering, and a hybrid approach. Two hypotheses were tested: (1) a pretrained transformer classifier can achieve at least 75% emotion classification accuracy on narrative text, and (2) a hybrid recommendation algorithm outperforms individual methods in preserving emotional alignment between stories and songs. Evaluation used 50 emotion-targeted autobiographical narratives, a curated catalog of 50 era-appropriate songs, and simulated profiles for 25 older adult users. The emotion classifier achieved 80% accuracy, exceeding the 75% threshold. The hybrid recommender achieved the highest recommendation score (0.941 versus 0.503 for random baseline), confirmed by paired statistical testing ( $p < 0.001$ ) and an ablation study across weighting parameters. These results establish a computational foundation for emotion-preserving music recommendation and motivate future clinical studies with older adult participants.

**Index Terms**—emotion detection, music recommendation, autobiographical memory, older adults, natural language processing, hybrid recommender systems

## I. INTRODUCTION

Over 61 million Americans are aged 65 and older, making them the fastest-growing demographic in the United States [1]. Social isolation affects over 25% of this population and is linked to cognitive decline and depression [2]. These challenges motivate the exploration of technology-supported interventions designed to promote emotional engagement and autobiographical memory recall.

Music has long been recognized as a powerful stimulus for memory retrieval. Janata et al. characterized music-evoked autobiographical memories (MEAMs) — vivid personal memories triggered by familiar music — and demonstrated their strong emotional associations [3]. Subsequent research showed that MEAMs occur naturally in everyday life and are experienced as vivid and emotional [4]. Recent work with healthy older adults aged 60–86 confirmed that music continues to

evoke rich autobiographical memories in later life [5], suggesting that music-based interventions could serve as therapeutic tools for this population.

Despite these findings, no prior system has systematically combined autobiographical narrative text with emotionally matched music recommendations. Current platforms such as Spotify optimize for preference similarity based on listening history and collaborative filtering, without accounting for the emotional content of personal memories or the therapeutic goal of preserving emotional alignment. The intersection of natural language processing (NLP), music information retrieval (MIR), and music therapy research presents an unexplored opportunity.

This study presents MelodyMemo, a two-module framework that (1) extracts emotions from autobiographical stories using a pretrained transformer model and (2) recommends emotionally aligned music using a hybrid algorithm. Two hypotheses are tested:

- **H1:** A DistilRoBERTa classifier can achieve  $\geq 75\%$  emotion classification accuracy across seven categories.
- **H2:** A hybrid recommendation algorithm outperforms individual methods and a random baseline on recommendation score.

This work makes three novel contributions: (1) applying transformer-based NLP to autobiographical memory text, a domain distinct from the social media corpora on which these models are typically evaluated; (2) creating a systematic emotion-to-musical-feature mapping bridging psychology and MIR; and (3) combining collaborative and content-based filtering with an explicit goal of emotion preservation rather than preference prediction.

## II. METHODS

### A. System Architecture

MelodyMemo consists of two sequential modules arranged in a pipeline (Fig. 1). A personal story enters **R1 (Emotion Extraction)**, which applies a pretrained transformer to detect the dominant emotion, assign a confidence score, and derive target musical attributes. These attributes flow into **R2 (Music Recommendation)**, which generates a ranked list of emotionally aligned songs.

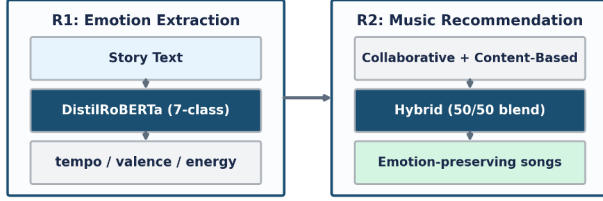


Fig. 1. MelodyMemo system architecture. Story text flows through R1 (emotion extraction) into R2 (music recommendation) to produce an emotion-preserving song list.

### B. R1: Emotion Extraction

Emotion classification uses DistilRoBERTa, a distilled variant of RoBERTa fine-tuned for emotion classification [6]. The model predicts softmax probability distributions across seven emotion classes: joy, sadness, anger, fear, surprise, love, and neutral. These categories follow the taxonomy in the model’s training data, broadly aligned with Ekman’s basic emotion framework [7]. Detected emotions are mapped to musical attributes (tempo in BPM, valence, and energy on 0–1 scales) using a literature-informed rule-based mapping.

### C. R2: Music Recommendation

R2 implements four algorithms: (1) a **random baseline** averaging 100 trials per user; (2) **collaborative filtering** using Jaccard similarity on liked-song sets to aggregate recommendations from similar users; (3) **content-based filtering** using cosine similarity between a user’s average feature vector and song features; and (4) a **hybrid** combining collaborative and content-based scores at equal weights ( $\alpha = 0.5$ ) with a consensus bonus for songs in both lists. This weighted hybrid approach is consistent with prior recommender systems literature [8]. The primary evaluation metric, recommendation score, is defined as the mean normalized ranking score of the top-5 recommended songs per user.

### D. Datasets

The evaluation used 50 autobiographical stories written from the perspective of older adults spanning seven emotion categories, a curated catalog of 50 era-appropriate songs (1939–1975) across 22 genres, and simulated profiles for 25 older adult users (ages 76–88) with 178 ratings and 85.8% matrix sparsity. A controlled synthetic dataset was used to ensure fixed ground-truth emotion labels and isolate model performance under controlled conditions. To assess classification consistency, DistilRoBERTa predictions were compared against GPT-4o and Gemini 2.5 Flash using identical seven-category classification tasks.

## III. RESULTS

### A. R1: Emotion Classification

The DistilRoBERTa model achieved 80.0% overall accuracy (40/50 stories), **exceeding the H1 threshold of 75%**. Table I

TABLE I  
DISTILROBERTA PER-EMOTION CLASSIFICATION RESULTS

| Emotion        | Stories   | Correct   | Accuracy     | Confidence   |
|----------------|-----------|-----------|--------------|--------------|
| Joy            | 7         | 7         | 100.0%       | 0.856        |
| Sadness        | 9         | 9         | 100.0%       | 0.977        |
| Anger          | 8         | 8         | 100.0%       | 0.919        |
| Surprise       | 6         | 6         | 100.0%       | 0.963        |
| Fear           | 8         | 7         | 87.5%        | 0.931        |
| Neutral        | 5         | 3         | 60.0%        | 0.822        |
| Love           | 7         | 0         | 0.0%         | 0.837        |
| <b>Overall</b> | <b>50</b> | <b>40</b> | <b>80.0%</b> | <b>0.907</b> |

presents the per-emotion breakdown. Five of seven categories achieved 87–100% accuracy: joy (100%), sadness (100%), anger (100%), surprise (100%), and fear (87.5%).

Two categories showed substantially lower accuracy: neutral (60%) and love (0%). All seven love stories were misclassified — four as joy, two as sadness, and one as fear — reflecting extensive lexical overlap between love and these categories. Fig. 2 visualizes the multi-model comparison and confusion patterns.

Cross-model comparison (DistilRoBERTa, GPT-4o, Gemini 2.5 Flash) was used to assess consistency: GPT-4o achieved 90% overall accuracy and Gemini 2.5 Flash achieved 88%, with both correctly classifying all love stories. All three models agreed on 70% of stories, and the majority vote confusion matrix shows strong diagonal dominance across most categories. This suggests that the love misclassification reflects a DistilRoBERTa-specific pattern rather than dataset ambiguity.

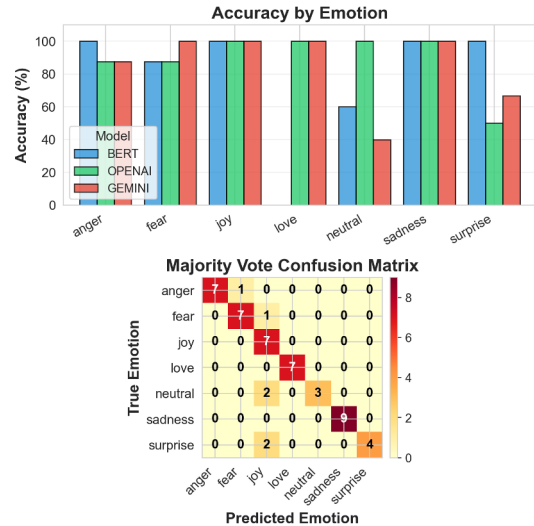


Fig. 2. Multi-model classification performance. Top: per-emotion accuracy across DistilRoBERTa, GPT-4o, and Gemini 2.5 Flash. Bottom: majority-vote confusion matrix.

TABLE II  
R2 ALGORITHM COMPARISON (25 USERS, TOP-5 RECOMMENDATIONS)

| Algorithm             | Score        | Emotion Match | Rating      | Genre Div. |
|-----------------------|--------------|---------------|-------------|------------|
| Random Baseline       | 0.503        | 19.1%         | 0.14        | 4.4        |
| Collaborative         | 0.607        | 21.6%         | 1.20        | 4.4        |
| Content-Based         | 0.893        | 36.0%         | 0.21        | 4.1        |
| <b>Hybrid (50/50)</b> | <b>0.941</b> | <b>34.4%</b>  | <b>0.64</b> | <b>4.4</b> |

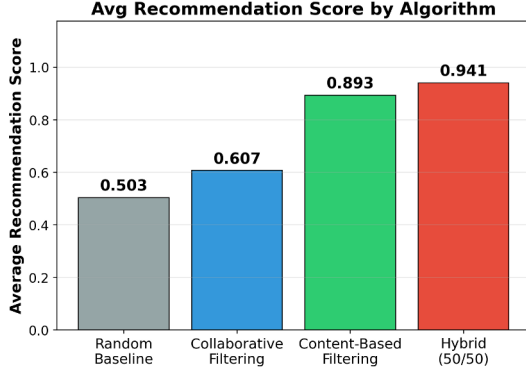


Fig. 3. Average recommendation score comparison across algorithms. The hybrid achieves the highest score.

### B. R2: Recommendation Performance

Table II summarizes the algorithm comparison across all 25 users. The hybrid algorithm achieved the highest average recommendation score (0.941), outperforming the random baseline by 87%, collaborative filtering by 55%, and content-based filtering by 5% (Fig. 3).

The results reveal complementary strengths that justify the hybrid design: content-based filtering achieved the highest emotion-match rate (36.0%) by explicitly matching musical features to the emotional query, while collaborative filtering achieved the highest average user rating (1.20) by leveraging community preference patterns. The hybrid algorithm captures both strengths, achieving the highest overall score while maintaining competitive performance across all metrics.

### C. Statistical Significance

Pairwise comparisons using both the Wilcoxon signed-rank test and paired t-test confirmed that the hybrid algorithm significantly outperformed all three alternatives on recommendation score (all  $p < 0.001$ ). An ablation study across 11 weighting ratios ( $\alpha = 0.0$  to  $1.0$ ) confirmed robustness, with recommendation scores remaining above 0.90 for all tested weights, **confirming H2**.

## IV. DISCUSSION AND CONCLUSION

MelodyMemo demonstrates that integrating transformer-based emotion extraction with hybrid music recommendation can preserve emotional alignment between autobiographical narratives and recommended songs. Both hypotheses were confirmed: the emotion classifier exceeded the 75% accuracy

threshold (H1), and the hybrid recommender significantly outperformed all alternatives with statistical significance (H2).

The error analysis reveals that emotion categories with distinct lexical signatures (anger, sadness, joy, surprise) are classified with high accuracy, while categories sharing overlapping vocabulary (love, neutral) remain challenging. The cross-model comparison against GPT-4o and Gemini 2.5 Flash suggests that larger context windows may help resolve these ambiguities, pointing toward fine-tuning on narrative-specific corpora as a promising improvement direction.

The primary limitation is the use of simulated user profiles and synthetic narratives rather than real participants. This study represents a controlled computational validation stage, providing a foundation for future evaluation with real older adult participants.

Future work will pursue two directions: (1) fine-tuning the emotion classifier on autobiographical narrative corpora to improve accuracy on challenging categories such as love and neutral; and (2) exploring configurable emotion strategies — where the system can match, cross-map, or gradually blend emotions depending on the therapeutic context — extending MelodyMemo from a fixed emotion-matching system to a flexible platform for personalized music-based interventions for older adults.

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